

Improper Disposal of Electronic Waste:

An E-Waste Segregation Awareness and Certified Recycler Directory Website

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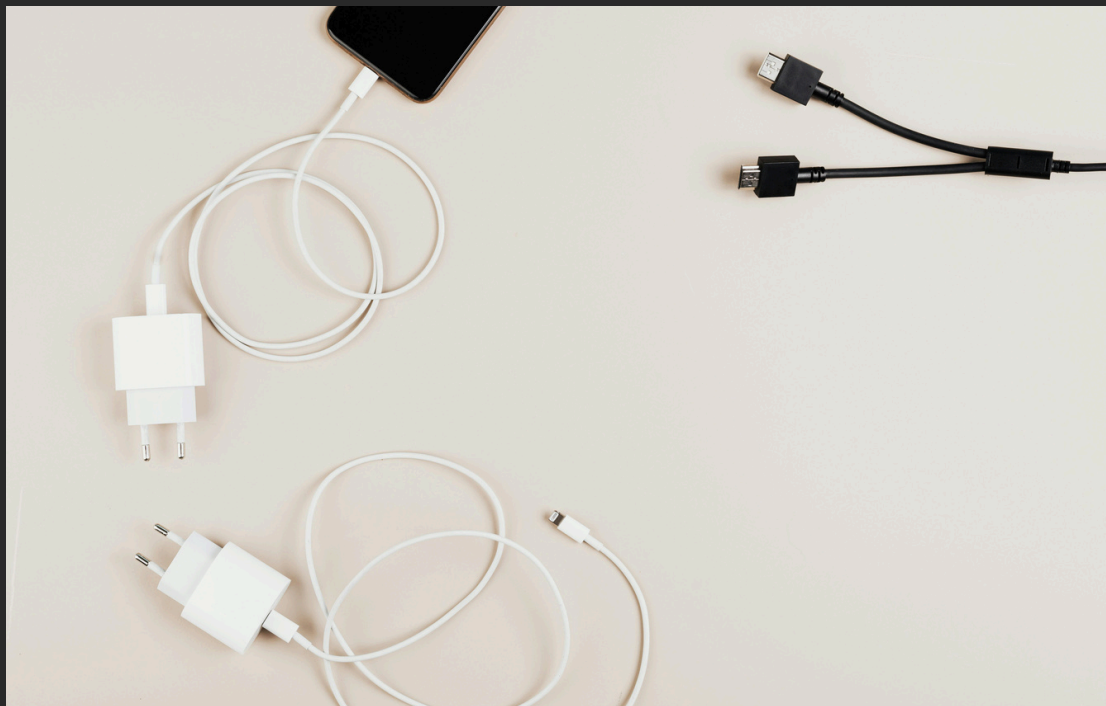
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What is E-waste ?



Why is improper e-waste disposal a problem?”

- Environmental Damage
- Health Risks
- Resource Loss
- Lack of Awareness

OUR SOLUTION

Identify E-Waste → Understand Its Category → Find an Authorised Recycler

What our website does:

- Identify e-waste
- Learn how to segregate it
- Find authorised recyclers
- Search by location

Causes of E-Waste

1

Rapid Tech Advancement

Frequent upgrades make devices obsolete fast

2

Short Product Lifespans

Planned obsolescence, cheap to replace

3

Rising Consumerism

Growing demand for phones, laptops, appliances

4

Falling Prices

Cheaper electronics drive higher turnover

5

Poor Repairability

Hard, costly fixes push people to dispose

6

Low Consumer Awareness

Few know how to recycle properly

7

Weak Collection Systems

Limited e-waste drop-off access

8

Corporate Upgrades

Businesses replace equipment regularly

Effects of E-Waste

Environmental

- Toxic metals leach into soil & water
- Air pollution from informal burning
- Valuable metals lost when unrecycled

Health

- Respiratory & neurological harm, cancer risk
- Greatest danger to informal workers, incl. children
- Heavy metals bioaccumulate in the food chain

Economic

- Loss of recoverable precious & rare metals
- High costs: pollution cleanup & health treatment
- Informal recycling sector often exploitative

Social

- Developing nations bear the import burden
- Health risks concentrated in low-income communities

Need for E-Waste Management

Prevent Contamination

Stops toxins entering soil & water

Recover Materials

Reclaims gold, silver, copper, rare earths

Cut Emissions

Less new-device manufacturing needed

Safer Recycling

Formal channels protect workers

Circular Economy

Supports reduce-reuse-recycle

Regulatory Compliance

Meets e-waste management rules

Protect Public Health

Shields vulnerable communities

Reduce Landfill Burden

Less soil & water pollution

Existing Solutions

How e-waste disposal is handled today — and where it breaks down

- **Informal Scrap Dealers**
- **Open Burning & Landfilling**
- **Scattered Govt. Records**
- **One-Way Awareness Drives**

Proposed Solution

E-Waste Segregation Awareness & Certified Recycler Directory Website

Part 1 — Segregation Guide

- **Helps a user identify which category their device falls into — small appliances, IT equipment, batteries, large appliances.**
- **Explains, in plain language, why that category needs careful handling.**

Part 2 — Recycler Directory

- **A searchable directory of certified, government-authorised recyclers, sourced from public CPCB / SPCB records.**
- **Filterable by state or city, and by the type of e-waste being disposed.**

Key Features

What the website offers, end to end

- **Interactive segregation guide with clear hazard explanations**
- **Searchable recycler directory, sourced from public PCB records**
- **Location / state filter to find a nearby recycler in seconds**
- **Short FAQ section addressing common disposal doubts**
- **Admin panel to keep the directory accurate over time**
- **Responsive layout — works on both mobile and desktop**

Website Overview & Design

Our website is designed to make responsible e-waste disposal simple and accessible. Main sections are :

-  **Home – Introduces the e-waste problem and the purpose of the portal**
-  **Segregation Guide – Helps users understand how different electronic devices should be handled**
-  **Find a Recycler – Helps users locate authorised e-waste recyclers**
-  **FAQ – Provides answers to common e-waste disposal questions**

Design approach:

Simple and user-friendly interface

Clear navigation between sections

Segregation Guide

The Segregation Guide helps users identify the correct way to handle their electronic devices.

How it helps:

- **Users select the category closest to their device**
- **Provides information about special handling requirements**
- **Explains why e-waste should not be placed in regular waste bins**
- **Guides users towards safe disposal**
- **Helps prevent hazardous materials from reaching the environment**

Example:

Old phone → Identify device → Check handling instructions → Send to an authorised recycler

Recycler Directory & Key Features

Key Features



Certified Recycler Directory

- **Helps users search for authorised recyclers**
- **Allows filtering based on location and type of e-waste**



Sort My Device

- **Helps users identify their device category**
- **Provides relevant disposal guidance**



Safe Disposal Guidance

- **Encourages users to hand over e-waste to authorised recyclers**

Platform Mechanics & User Journey

Step 1: Interactive Segregation Guide ("Which bin does it belong in?")

- Categorizes devices by toxic handling groups (smartphones, batteries, IT equipment, household appliances).
- Outlines specific environmental hazards alongside pre-disposal preparation steps (e.g., data wiping, battery detachment).

Step 2: Certified Recycler Directory

- Searchable database filterable by State and E-Waste Type.
- Synchronized with public Pollution Control Board (CPCB/SPCB) authorization lists to guarantee legitimate drop-off points.

Strategic Design & Feasibility

1. Frictionless Consumer Experience:

- Eliminates pickup coordination, account creation, or reward tracking in favor of instant, 2-click information discovery.

2. Zero-Inventory / Low Overhead Model:

- Operates without managing transport logistics, warehouses, or fleets, enabling low-cost deployment and rapid scaling.

3. Regulatory Alignment:

- Directly aligns with India's E-Waste (Management) Rules, 2022 mandates for consumer awareness and public record accessibility.

Circular Economy & Sustainability Value

1. Hazardous Material Containment:

- Prevents Lead (Pb), Mercury (Hg), and Cadmium (Cd) from leaching into soil, groundwater, and urban ecosystems.

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2. Urban Mining & Resource Efficiency:

- Facilitates the extraction and reuse of gold, silver, copper, and rare earth elements, reducing reliance on virgin mining.

3. UN Sustainable Development Goals (SDGs):

- SDG 3: Good Health & Well-Being
- SDG 6: Clean Water & Sanitation
- SDG 11: Sustainable Cities & Communities
- SDG 12: Responsible Consumption & Production
- SDG 13: Climate Action

IMPACT

Environmental Impact

- Encourages responsible e-waste disposal
- Helps reduce improper dumping and burning
- Supports safer handling of hazardous components

Social Impact

- Makes e-waste information easier to understand
- Connects awareness with a practical next step

Sustainability Impact

- Supports responsible consumption and recycling awareness
- Connects with SDGs 3, 6, 11, 12 and 13

CONCLUSION

Second Circuit turns e-waste awareness into a simple action pathway:

**UNDERSTAND → SEGREGATE → FIND AN AUTHORISED RECYCLER → DISPOSE
RESPONSIBLY**

The website demonstrates how a focused digital solution can encourage better e-waste management while remaining simple, accessible and practical.

Thank You